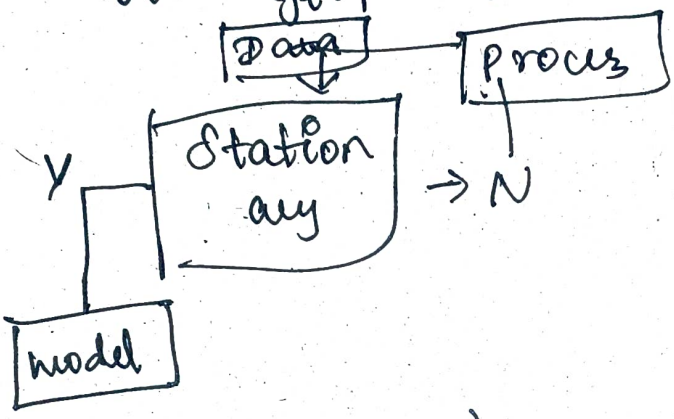
06/10/26

Stationary & non-stationary:-

A stochastic process is said to be stationary if its mean & variance are constant over time & value of covariance b/w two time period depends on the distance or lag b/w two time periods & not on the actual time at which the covariance is computed

$\Delta y_t = \delta \rightarrow$ unit root $y_{t-1} + \varepsilon_t$



Random walk (R-w):-

(Behaviour of stock prices)

Suppose $\{\varepsilon_t\}$ is purely random series with mean μ & variance σ^2. Then the process $\{x_t\}$ is said to be RW if

$$x_t = x_{t-1} + \varepsilon_t$$

Let us assume $x_0 = 0$: then

$$x_1 = \varepsilon_0$$

$$x_2 = x_1 + \varepsilon_2 = \varepsilon_1 + \varepsilon_2 \text{ & } \ldots$$

$$\text{So } x_t = \sum_{t=1}^{t} \varepsilon_t \Rightarrow E(x_t) = t\mu$$

$$\text{Var}(x_t) = t\sigma^2$$

$$y_t = \alpha y_{t-1} + u_t$$

$$y_t - y_{t-1} = \alpha y_{t-1} - y_{t-1} + u_t$$

$$\Delta y_{t-1} = (\alpha - 1) y_{t-1} + u_t$$

$$\Delta y_{t-1} = \delta y_{t-1} + u_t$$

$\boxed{\delta = 1}$ → DICKEY FULLER Test

Stationary.

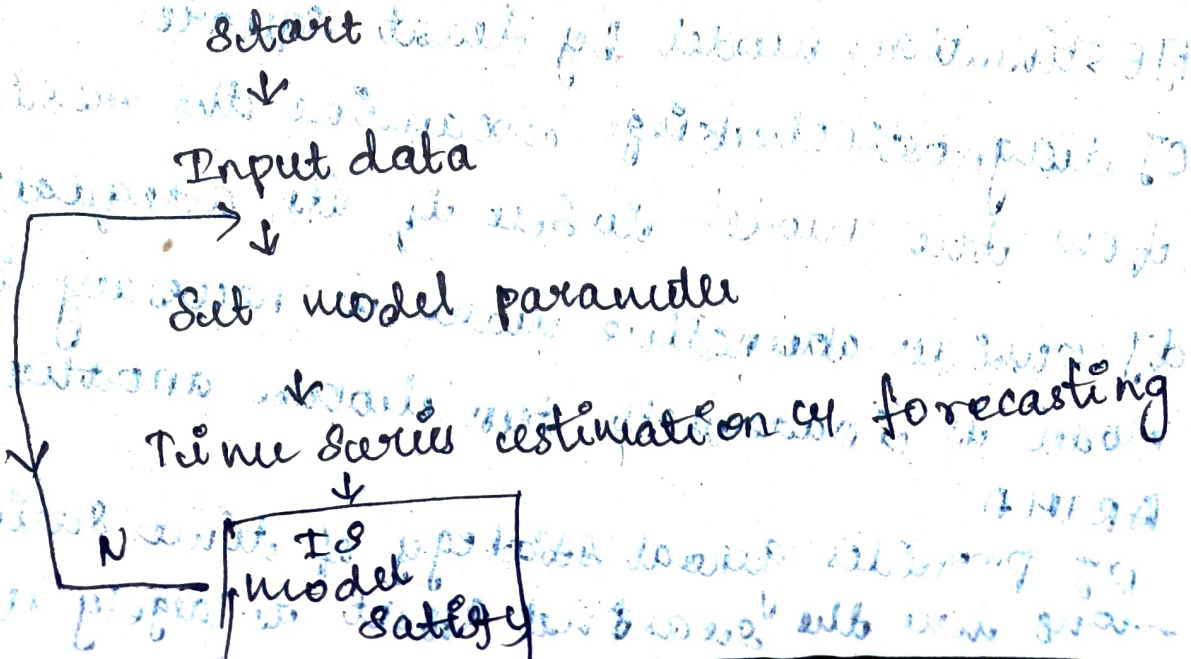
Graphical → non stationary (mean, variance)

Numerical → stationary.

08/10/26

ARIMA Model:- Flexible

It is a model method helps to understand complex data, predict future points, essentially part of BJ approach.



↓ Y

Output data

↓

Stop

The general non-seasonal model is known as ARIMA(P,d,q)

where,

P → order of auto regressive part

d → degree of differencing involved

q → order of moving avg part this class of model is extended to non-stationary time-series by allowing to difference the data.

Box-Jenkins model:-

It is based on fitting an ARIMA model to a given set of data then taking conditional expectation

Steps:-

a) model identification: examine the data to see which model is fits to

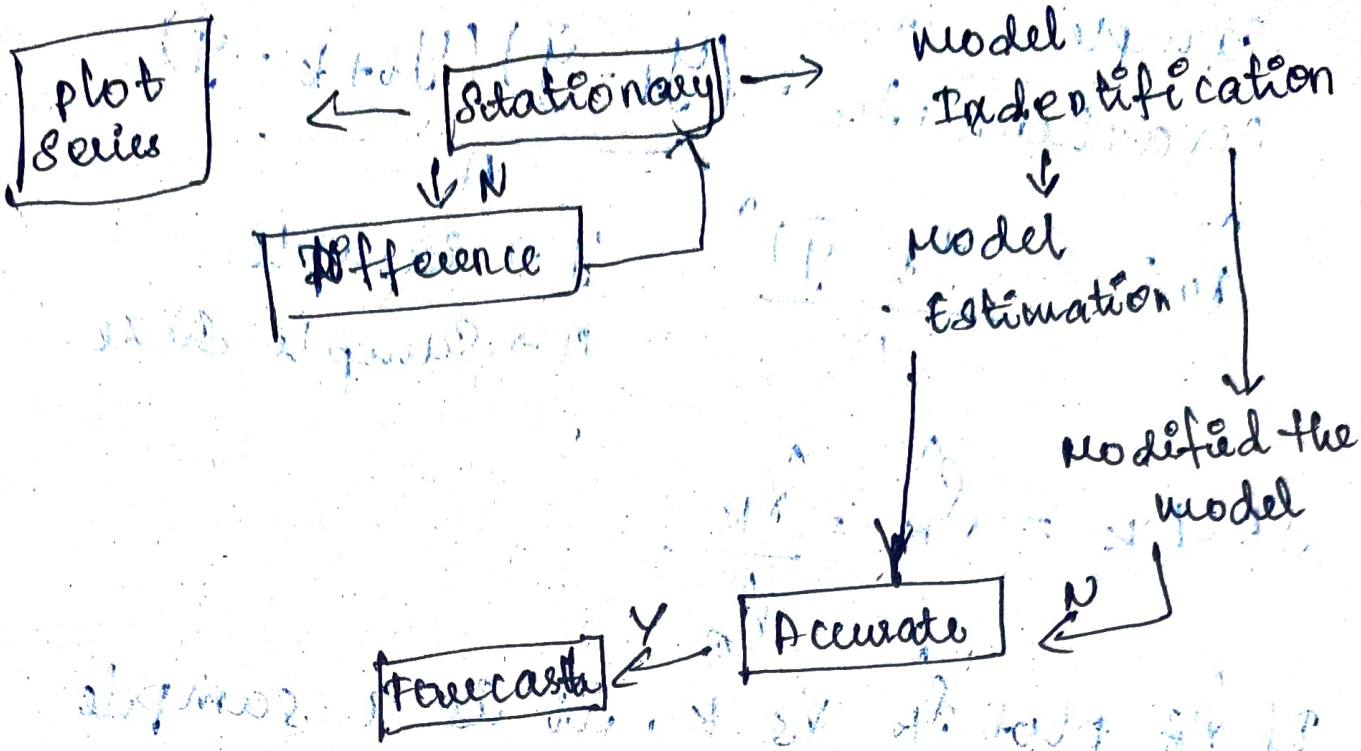
b) Estimation: model by least square.

c) diagnostic checking: examine the residual from the model to see if its adequate.

d) consider alternative model is necessary if the model is inadequate then choose another model.

ARIMA

BJ provides general strategy of time series forecasting where the seasonal effect is largely removed



y_t	Δy_t	$\Delta^2 y_t$	$\Delta^3 y_t$
23	-	-	-
25	2	-	-
27	2	0	-
32	5	3	3
33	1	-4	-7
37	4	3	7
42	5	1	-2
45	3	-2	-3

ARIMA (p, d, q)
(1, 0, 1)

$\Rightarrow \alpha_0 + y_{t-1} + \varepsilon_{t-1} + \varepsilon_t$

13/10/26

Correllogram:-

A sample autocorrelation for has to be completed for which sample covar at lag k of sample var has to be completed

Sample covariance: $\hat{\gamma}_k = \frac{\sum (y_t - \bar{y})(y_{t+k} - \bar{y})}{n}$

$\hat{\gamma}_0 = \frac{\sum (y_t - \bar{y})^2}{n}$ $\bar{y} \rightarrow$ sample

$n \rightarrow$ sample size.

$SACF_k = \hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$

If we plot $\hat{\rho}_k$ vs k, it is a sample correllogram.

white noise: seq of random disturbance with 0 mean & constant var

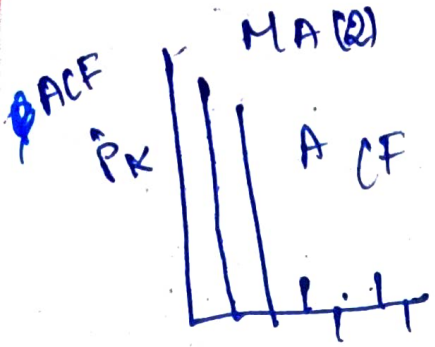
⇒ good forecast ⇒ residual ⇒ white noise

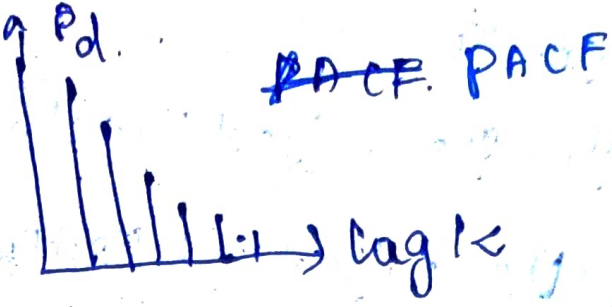
TO compute sample autocorrelation fn $SACF(\hat{\rho}_k)$. compute sample cov $\hat{\gamma}_k$ & sample variance $\hat{\gamma}_0$

∴ sample covar at lag k by $n-k$ &

Sample var by $n-1$

$\hat{\gamma}_k = \frac{\sum (y_t - \bar{y})(y_{t-k} - \bar{y})}{n}$

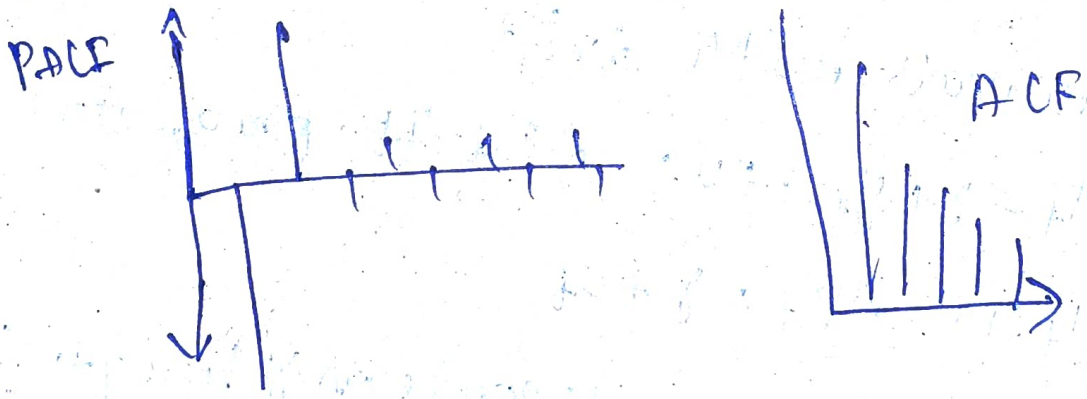




Model	ACF	PACF
AR(1)	Tails off	cuts after lag 1
AR(2)		
MA(1)	cut of lag 1	Tails off
MA(2)	cut of lag 2	Tails off
ARMA(P,q)		

AR(2) ⇒

$$y_t = \alpha_0 - \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \varepsilon_t$$



11/10/26

Random walk with drift

① $y_t = y_{t-1} + v_t$ $\begin{cases} v_t \text{ white noise mean } 0 \text{ var } \sigma^2 \\ v \sim N(0, \sigma^2) \end{cases}$

$\Rightarrow y_1 = y_0 + v_1$

$y_2 = y_1 + y_2 = y_0 + v_1 + v_2$

$Y_3 = y_2 + v_3 = y_0 + v_1 + v_2 + v_3$

$\Rightarrow y_t = y_0 + \Sigma v_t$

$\therefore E(y_t) = E(y_0 + \Sigma v_t) = y_0$

Hence RWM without drift is non-stationary stochastic process

① can be written as

$$y_t - y_{t-1} = v_t$$

$$\Delta y_t = v_t$$

Hence y_t is non-stationary & Δy_t is stationary

So First diff of RWM is stationary

II. Random walk with drift

$$y_t = \gamma + y_{t-1} + v_t \quad \gamma \rightarrow (\text{drift parameter})$$

So $y_t - y_{t-1} = \Delta y_t = \gamma + v_t$

y_t drifts up / down depends on γ being +ve/-ve

$$E(Y_t) = y_0 + t\gamma$$

$$Var(Y_t) = t\sigma^2$$

Eq ① $\Rightarrow y_t = \rho y_{t-1} + v_t \quad -1 < \rho < 1$

(Markov 1st Order AR Model)

(a) If $\rho = 1 \Rightarrow$ RW without drift i.e unit Root problem / non-stationary

(b) If $|\rho| < 1 \Rightarrow$ stationary

If trend is completely predictable $\Rightarrow$ Deterministic trend

If not predictable $\Rightarrow$ Stochastic trend

(*) $y_t = \beta_1 + \beta_2 t + \beta_3 y_{t-1} + v_t$



Pure random walk:-

$\beta_1 = 0, \ \beta_2 = 0, \ \beta_3 = 1$

$y_t = y_{t-1} + v_t$

$\Delta y_t = (y_t - y_{t-1}) = v_t$ → without drift

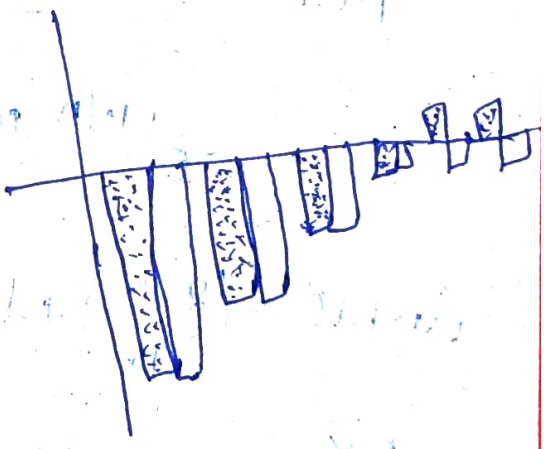
Random walk with drift:-

$\beta_1 \neq 0, \ \beta_2 = 0, \ \beta_3 = 1$

$y_t = \beta_1 + y_{t-1} + v_t$

$(y_t - y_{t-1}) = \Delta y_t = \beta_1 + v_t$

$y_t = \beta_1 + \beta_2 t + \beta_3 y_{t-1} + v_t$



Deterministic trend:-

$\beta_1 \neq 0, \ \beta_2 \neq 0, \ \beta_3 = 0$

$y_t = \beta_1 + \beta_2 t + v_t$

Random walk with drift or Deterministic trend:-

$\beta_1 \neq 0, \beta_2 \neq 0, \beta_3 = 1$

$y_t = \beta_1 + \beta_2 t + y_{t-1} + v_t$

$\Delta y_t = \beta_1 + \beta_2 t + v_t$

20/10/26

$$SIC = n^{k/n} \cdot \frac{\sum \hat{u}^2}{n} = n^{k/n} \cdot \frac{RSS}{n}$$

$$\ln SIC = \frac{k}{n} \ln(n) + \ln\left(\frac{RSS}{n}\right)$$

$\frac{k}{n} \ln(n) \rightarrow$ penality factor

SIC imposes harsher penality then AIC lower value of SIC is better. It can be used to compare in sample or out of sample forecasting performance of a model.

$$AIC = e^{2k/n} \frac{\sum \hat{u}_i^2}{n}$$

$k \rightarrow$ no. of Exogeneous variable

$$= e^{2k/n} \frac{RSS}{n}$$

$n \rightarrow$ no. of observation

$$\ln AIC = \left(\frac{2k}{n}\right) + \ln\left(\frac{RSS}{n}\right)$$

$\frac{2k}{n}$ - penality factor

Compare 2 models. Model with lower AIC is preferred.

ministic

lowest
to compare
ing

rogeneous

Stevation

e AIC ds

① $y = \alpha_0 + \alpha_1 x + u_t$

② $y = \alpha_1 x + u_t$

③ $y = \alpha_0 + \alpha_1 x + \alpha_2 t + u_t$

④ $y = \alpha_1 x + \alpha_2 t + u_t$

21/10/26

B-J method :-> method

Identifies → Estimation → Diagnostic checking → Forecast

Ide the past data w which model is suitable

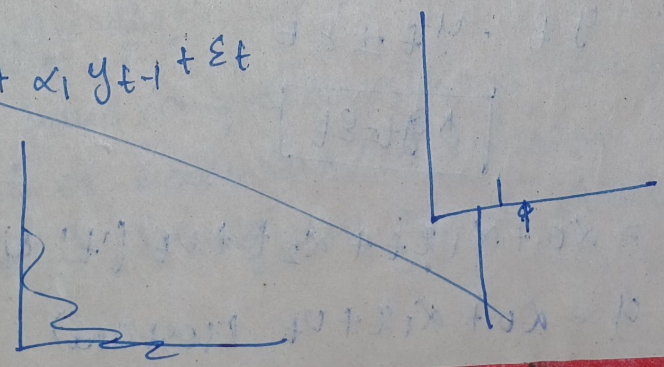
Model parameters

chens

y_t	Δy_t	$\Delta^2 y_t$
1100	-	-
1120	20	-
1133	13	-7
1149	16	3
1150	1	-15

Uses:-

$y = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$



① $y = \alpha_0 + \alpha_1 x + u_t$

② $y = \alpha_1 x + u_t$

③ $y = \alpha_0 + \alpha_1 x + \alpha_2 t + u_t$

④ $y = \alpha_1 x + \alpha_2 t + u_t$

21/10/26

B-J method :--> method

Identifies → Estimation → Diagnostic checking → Forecast

Identifies the past data in which model is suitable

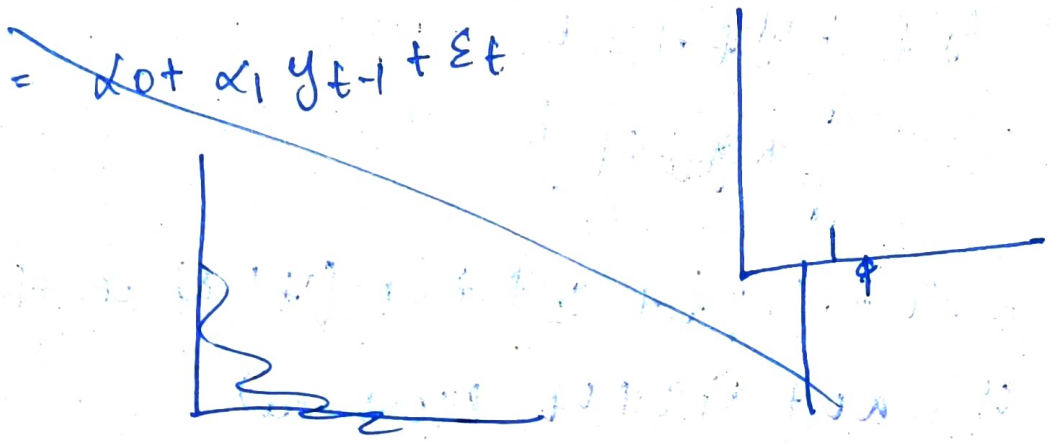
Estimation: Model parameters

Diagnostic checking: check

y_t	Δy_t	$\Delta^2 y_t$
1100	-	-
1120	20	-
1133	13	-7
1149	16	3
1150	1	-15

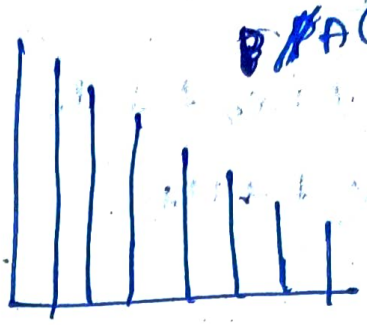
Uses:-

$y = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$

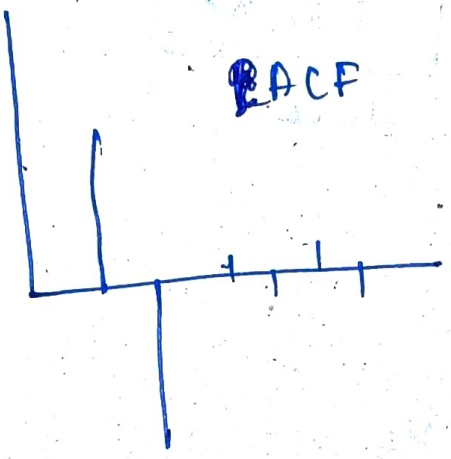


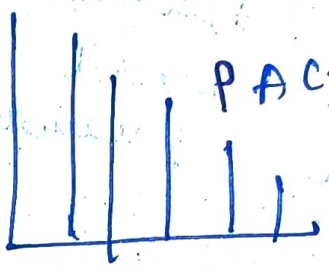
$y = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$





$y_t = \alpha_0 + \alpha_1 \varepsilon_{t-1} - \alpha_2 \varepsilon_{t-2} + \varepsilon_t$





22/10/26

stationary — graph
math.

without drift → no constant

Random walk with constant ~~$y_t = y_{t-1} + \varepsilon_t$~~.
without constant.

$y_t = y_{t-1} + \varepsilon_t \sim N(0, \sigma^2)$

$y_t - y_t = \varepsilon_t$

$\boxed{\Delta y_t = \varepsilon_t}$

$y_t = \alpha_0 + \alpha_1 x_t + \alpha_2 t + v_t$ ('t' as const)

$y = \alpha_0 + \alpha_1 x + v_t$ (constant)

③ $y = \alpha_0 x + \alpha_1 t + u_t$ (only t)

④ $y = \alpha_0 x + u_t$ (no t & no c)

$y_t = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$

$y_t = \alpha y_{t-1} + \varepsilon_t$ (No, c & t)

$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 t - 1 + \varepsilon_t$ (with t & c)

$y_t = \alpha_1 y_{t-1} + \alpha_2 t + u_t$ (t)

Note

$|P| < 1 \rightarrow$ not stationarity
↳ unit root problem.

23/10/26

ARIMA(1,1,1) = ARIMA(1,0,1)

$= \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 \varepsilon_{t-1} + u_t$

ARIMA(1,0,0)

$\Delta y_t = \alpha_0 + \alpha_1 \Delta y_{t-1} + \varepsilon_t$

ARIMA(0,0,1)

$y_t = \alpha_0 + \varepsilon_t + \theta_1 \varepsilon_{t-1}$

$y_t = \alpha_0 + \alpha \varepsilon_{t-1} + u_t$

ARIMA(0,1,1)

$\Delta y_t = \alpha_0 + \alpha_1 \Delta \varepsilon_{t-1} + u_t$

Dickey - Fuller test :- ~ I

Authors have computed critical Tau statistics on the basis of monte carlo simulation

If $\delta = 0$ is is rejected $\Rightarrow$ Ts is stationary

$\beta_1 \neq 0$, $\beta_2 = 0$; $\beta_3 = 1$
$\beta_1 \neq 0$; $\beta_2 = 0$, $\beta_3 = 1$
$\beta_1 = 0$; $\beta_2 = 0$; $\beta_3 = 1$
} Identify ew

$y_t = \beta_1 + \beta_2 t + \beta_3 y_{t-1} + u_t$

unit root ① $\rightarrow y_t = \rho y_{t-1} + \varepsilon_t$ → white noise

where $\rho = 1$; UR ≠ NS ≠ RW ① $-1 \leq \rho \leq +1$

Should not reg y_t on y_{t-1} ⇒ Biased

① $y_t - y_{t-1} = \rho y_{t-1} - y_{t-1} + \varepsilon_t$

$\Delta y_t = (\rho - 1) y_{t-1} + \varepsilon_t$

$\Delta y_t = \delta y_{t-1} + \varepsilon_t$ $H_0: \delta = 0$

$H_1: \delta < 0$

when $\delta = 1$, $\Delta y_t = \varepsilon_t$ (st : u)

2 Feb 26

ARIMA:-

* It is an iterative process
* It is a form Box Jenkins model
* checking whether its stationary or non stationary using the level of I
* Superior to exponential smoothing method
* used for data sample greater than 40
* provides a perfect fit for a data provided
tried.

unit root:-

$y_t = \delta y_{t-1} + \varepsilon_t$

$y_t - y_{t-1} = \delta y_{t-1} - y_{t-1} + \varepsilon_t$

$\Delta y_t = (\delta - 1) y_{t-1} + \varepsilon_t$

$\Delta y_t = \delta y_{t-1} + \varepsilon_t \quad -1 \leq \delta \leq +1$

if $\delta = 1$ unit root R.W non stationary

if $\delta = 0$ i.e there is a unit root or time series is non stationary)

H1: $\delta < 0$ i.e time series is stationary)

Dickey Fuller test

① with trend

② with intercept w trend

③ NO intercept w trend

case 1: $y_t = y_{t-1} + \varepsilon_t \quad \Delta y_t = \varepsilon_t$

Differential stationary process

case 2: $y_t = \beta_1 + y_{t-1} + \varepsilon_t \quad \Delta y_t = \beta_1 + \varepsilon_t$

Stochastic trend

case 3: $y_t = \beta_1 + \beta_2 t + \varepsilon_t \rightarrow$ Trend stationary proces

cas 4: $y_t = \beta_1 + \beta_2 t + y_{t-1} + \varepsilon_t \quad \Delta y_t = \beta_1 + \beta_2 t + \varepsilon_t$

with drift & 't'

case 5: $y_t = \beta_1 + \beta_2 t + \beta_3 y_{t-1} + \varepsilon_t$

Trend stationary process AR(1)

Check whether stationary or not.

⇒ Graph → Not mean reverting

⇒ Correlogram → ACF, PACF

⇒ Dickey fuller unit Root → $\delta = 1 \Rightarrow$ NS

Detrending

⇒ Error of plot

⇒ First difference

⇒ MA - Moving average.

27/10/26

ARIMA B-J method ACF-PACF

Un-3

⇒ Stationary & NS

⇒ Unit Root

⇒ Dickey Fuller

⇒ Augment DF

⇒ Cointegration

3 Specification of DF test :-

① y_t is a RW : $\Delta y_t = \delta y_{t-1} + \varepsilon_t$

② y_t is RW with drift : $\Delta y_t = \beta_0 + \delta y_{t-1} + \varepsilon_t$

③ y_t is RW with trend & drift : $\Delta y_t = \beta_0 + \beta_1 t + \delta y_{t-1} + \varepsilon_t$

ε_t is white noise

$H_0: \delta = 0$ i.e y_t non stationary

$H_1: \delta < 0$ y_t stationary

If H_0 is refuted y_t is Stationary

τ-test statistics

(If tau value is smaller than critical tau) value ↓ Reject H0.

DF :- $\Delta y_t = \delta y_{t-1} + \varepsilon_t$ → H0: $\delta = 0$ - non stat, H1: $\delta < 0$ → stati

DF : $\Delta y_t = f(y_{t-1})$

DF Assumes: ε_t is white noise

→ AIC / SIC / BIC (ADF) → For taking lag.

→ Akaike / Schwarz & others info criteria.

→ In ADF we still check $\delta = 0$.

DF : $\Delta y_t = f(y_{t-1})$

ADF : $\Delta y_t = f(y_{t-1}, \Delta y_{t-1})$ } VR (Unit Root) → Akaike will help to take lag.

(PP) - Phillip Perron (PP) unit root test

PP uses non parametric statistical method to take care of serial correlation in error term without adding lag.

DF :- ① $\Delta y_t = (y_{t-1})$

② $\Delta y_t = (y_{t-1}, t)$

③ $\Delta y_{t-1} = (y_{t-1}, t, \alpha)$

Co integration :-

	Income	cons
1st diff	~	~
2nd diff	~	~
3rd diff	~	~

We should use same diff for both features, even if one of the data get stationary before than another one.

I(0) $y_t = \alpha_0 + \alpha_1 x_t + u_t$

I(1) $\Delta y_t = \alpha_0 + \alpha_1 \Delta x_{t-1} + u_t$ ⇒

α_1 → cointegration parameter

cointegration eq.

cointegration → means long run relationship

29/10/26

6- integration:

* 2 series share the same common trend
* 2 variables are cointegrated if they have a long term equilibrium relationship b/w them
* If distance b/w 2 variables are equal, y_t & x_t are cointegrated
* I(1) series are cointegrated if there exist atleast one linear combination of these variables that is stationary (same probability dist)
* Granger A test for cointegration can be thought of a pre-test to avoid Spurious Regression situation

cointegration:

$y_t = \alpha_0 + \alpha_1 L(x_t) + u_t$

u_t is constant

$y_t = \alpha_0 + \alpha_1 L(x_t)$

D.F & A.D.F:-

① Data collection

② checking stationarity

③ D.F: $\Delta y_t = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$

if the white noise is correlated

④ ADF: $\Delta y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 \Delta y_{t-1} + \varepsilon_t$

Error - correction Model $\Rightarrow \Delta \varepsilon_t = P(\varepsilon_{t-1})$

Granger - casuality Model:-

$$y_t = f(y_{t-1}, x_{t-1})$$

$$x_t = f(y_{t-1}, x_{t-1})$$

Vector Auto regressive Model (VAR): Mutual relationship b/w 3 variable

$$y_t = f(x_{t-1}, y_{t-1}, z_{t-1})$$

$$x_t = f(x_{t-1}, y_{t-1}, z_{t-1})$$

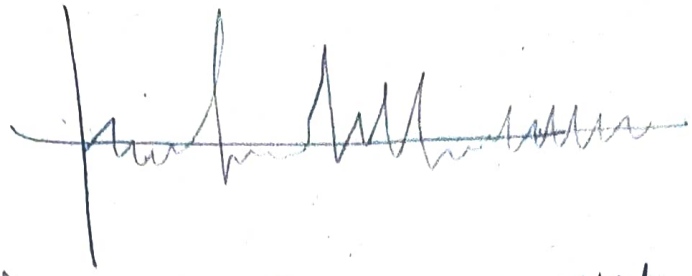
$$z_t = f(x_{t-1}, y_{t-1}, z_{t-1}).$$

Note: No correlation b/w error terms.

2/1/26

ARCH(P)

Yesterday's unexpected news may influence today volatility.



* Large price movement tend to be followed by large movement & small movement tend to be fol by small movement

Note
volatility clustering

Return affects Investment

Sunday 100

Monday 110 FOMO

$\frac{110-100}{100} = 10\%$

ARCH(1) ..... ARMA (1,0,0)

$x_t^2 = \alpha_0 + \alpha_1 x_{t-1}^2 + \varepsilon_t$ → ARCH model.

↳ var is a function of lag e^2

$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$

If α_1 is +ve (Today → Previous high) then is a volatility clustering

$\alpha_1 = 0$ → (NO volatility & clustering)

var → variance

ARCH model Var = f(lag e^2)

GARCH model Var = f(lag e^2, Var $t-1$)

↳ Both model used for volatility clustering